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**Advancing Clean Energy
Solutions for Climate-
Resilient and Sustainable
Development in Nigeria**

Design of a solar-powered water pumping system for sustainable irrigation in rural communities

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Abstract

This paper presents the design of a photovoltaic (PV) water pumping system for the purpose of sustainable irrigation in crop production. The system was to draw water from a well with a dynamic level of 9.7 m and a total dynamic head (TDH) of 15.7 m at a flow rate of 2.4 m³/hour. A power pump of 258 W was obtained from the pump performance curve, which was required to efficiently extract water from the well and discharge it to an overhead tank. The study then employs the HOMER software to evaluate the feasibility, performance, and economic viability of a solar-powered water pumping system in comparison to a conventional diesel-driven system. The results obtained clearly show that the designed solar water pumping system solution was able to produce the energy required for irrigation. The economic viability of the system was demonstrated by its Net Present Value (NPV) of ₦1.8 million, Levelized Cost of Electricity (LCOE) of ₦184.17/kWh, Payback Period (PBP) of 0.7 years, and Internal Rate of Return (IRR) of 158.7%, which were determined by actual market prices and local inflation factors. Additionally, this study demonstrates that installing the solar system could stop the release of 102 kg of CO₂ per year. This would also cut diesel fuel use by 38.9 L.

Keywords

Solar Power, Crop Irrigation, Solar Pump, Solar Panel, Water Tank, Nigeria

1. Introduction

Agriculture is the lifeblood of human civilization, providing sustenance and livelihoods to billions worldwide. Crop irrigation is vital throughout the world in order to provide the world's ever-growing population with enough food (USGS, 2018). Irrigation is the controlled application of water for agricultural purposes to provide water needs that are not met by rainfall. In Nigeria, a large percentage of land is devoted to rain-fed production, while a small percentage of land is irrigated (FAO, 2020). Nigeria's rain-fed agricultural systems will need better irrigation systems to improve crop yields since most farmers resort to manual means of crop irrigation when there is no rain. More frequent droughts and water shortages

represent a big risk to livelihoods and food security, especially for the most vulnerable populations in rural areas.

Solar power solutions are emerging in agriculture to make the industry more sustainable in the future (Ani, 2025). Many successful SPIS have been implemented in different regions and for a variety of farming practices. For example, they have been adopted by large-scale commercial farms, such as the Bowery Farming facility in New Jersey, United States. Bowery Farms has been able to grow a variety of crops year-round using solar panels and advanced hydroponic technology. Additionally, the farm's carbon footprint is reduced, making it a sustainable and environmentally friendly operation. Another successful case study is the Solar Pump Irrigators' Cooperative Enterprise (SPICE) in India which is a community-based organization that helps smallholder farmers use solar irrigation systems. SPICE has helped farmers switch from diesel-powered pumps to solar-powered systems by offering financing options and technical support. Farmers have been reporting increased crop yields, reduced operational costs, and better access to water. In the Karshi community, a significant percentage of the population earns their living from agriculture. Many small-scale farmers work all day to provide food for their own families and selling produce when there is surplus. These farmers struggle to obtain enough water to irrigate the crops, thus a more sustainable solution for small-scale farmers' irrigation systems will focus on using reliable and clean solutions.

Previous solar water pumping systems proposed in literature, faced significant shortcomings related to high initial capital costs, weather-dependent performance, and system reliability. Many projects failed due to "static" rather than "dynamic" design, ignoring real-time fluctuations in solar input and water demand, leading to inefficient operation. These historical challenges highlight the need for this research.

Significance, Aim, Objective, and Novelty of the study

Solar-powered water pumping systems for sustainable irrigation is crucial for rural communities because it provides a cost-effective, reliable, and eco-friendly alternative to diesel pumps, enabling year-round food production and boosting economic resilience. It transforms agriculture by eliminating expensive fuel costs and reducing the carbon footprint of farming in remote area. The aim of this research is to design a solar-powered water pumping system that is effective for farm irrigation systems. This will be achieved by employing the HOMER software as a simulation tool to determine the optimal configuration. Its objective is to determine the more cost-effective option: a solar-powered pumping system or a diesel-powered pumping system. This study is applicable and repeatable, providing information to developers, legislators, and local farmers who are considering utilizing solar water pumping systems for sustainable irrigation. In this paper, we suggested a deferrable load method for electrical load modelling. A deferrable load is an electrical load that requires a certain amount of energy within a given time period, but the exact timing is not important; it can wait until power is available. When a load is connected to storage, it is typically categorized as deferrable. The novelty of the study is that the use of deferrable load in the design of a solar-powered water solution for sustainable agriculture has not been done before in Nigeria.

2. Literature Review

A number of papers have been written about solar PV water pumping systems, and several studies have been conducted on sizing techniques and optimization aspects of solar-powered water pumping systems. All these papers use the same method of pump sizing which mainly depends on the flow rate of water and the dynamic water head, but they use different methods of sizing the PV array which mainly depends on the efficiency of PV modules and the quality and quantity of solar irradiance available in the location of water source (Almarshoud, 2016). The reviews by Sergio and Rousse, (2021) show that the proposed methods for sizing are very similar in most of the published articles. System sizing is one of the most important steps in determining the solar-powered water pumping system's long-term performance. In other words, a mistake in sizing can lead to poorer system performance, by which the system is not capable of pumping enough water when confronted with expected consumption (Wenham, 2007). Grant et al., (2022) presented the solar-powered drip irrigation optimal performance model to optimize system designs. Their findings show that solar-powered drip irrigation could work in real-world conditions; and could make solar-powered drip irrigation more accessible to smallholder farmers. Sergio and Rousse, (2021) looked at the scientific literature on solar-powered water pumping systems and found that Pande et al., (2003) proposed several criteria for designing a PV pumping system for crop irrigation. First, water requirements are determined for each plant, as well as the theoretical energy needed to pump water according to these requirements, taking into account pressure losses. In terms of water consumption, many variables are mentioned, such as the phases of plant growth, soil type and season. Also, the consumption pattern varies over time throughout the year. However, only the peak consumption is considered when designing the system. The selection criteria for a pump are the volume of water to be pumped, the working pressure of the system, the pressure drops and the efficiency of the pump-motor assembly. However, the process of selecting the pump is not explicit. The number of PV modules is based on the theoretical efficiency of the pump-motor unit, the nominal power of the modules and the rated power of the pump. These criteria ultimately showed key development milestones for future solar-powered water pumping systems.

3. Methodology

The methodology for the proposed solar-powered pumping system is described in the following sections.

3.1. Solar Energy Potential and Insolation

Karshi is a satellite town in the Abuja Municipal Area Council (AMAC), one of six council areas of the Federal Capital Territory (FCT) (Ojo et al., 2021). The Karshi area council is located south-east of the FCT, and has a boundary with Nasarawa State (Felix et al., 2024). The Karshi Area of Abuja is shown in Figure 1. Karshi falls within Latitude $8^{\circ}49'40''$ North and Longitude $7^{\circ}33'0''$ East (Longitude-Latitude-Maps, 2024; The GPS Coordinates, 2024). It is about 50 km from Abuja city centre (Mohammed et al., 2016). Karshi observes three weather conditions per year. This includes the rainy season, which is warm and humid, and the dry season, which is very hot and dry. In between these seasons, there is a short period of

harmattan (December to January) accompanied by the north-east trade wind, featuring dust haze, intensified coldness, and dryness (Ajibade et al., 2021). There is also a period of dust-free weather (November and February); which is known for its high irradiance and clear weather conditions, while March marks the transition from the dust-free period of February to the rainy season. The wet season, which starts in April and ends in October, experiences its peak rainfall in June and September. Karshi experiences cooler conditions during the rainy season, even though it has high temperatures on average. Karshi weather conditions are very good for agricultural products such as tubers, root crops, grain, and vegetables. These farmers are involved in both rain-fed agriculture and irrigation farming. Farming activities are undertaken on a smallholder basis, with per capita land ownership averaging about 0.5 ha (Safirat et al., 2023). Karshi is still largely underdeveloped, despite its linkage to the Abuja main town (Okezie et al., 2022). However, the area is predominantly rural and mainly consists of indigenous communities who are mostly involved in farming and related activities. Table 1 shows the monthly average of daily insolation for this study.

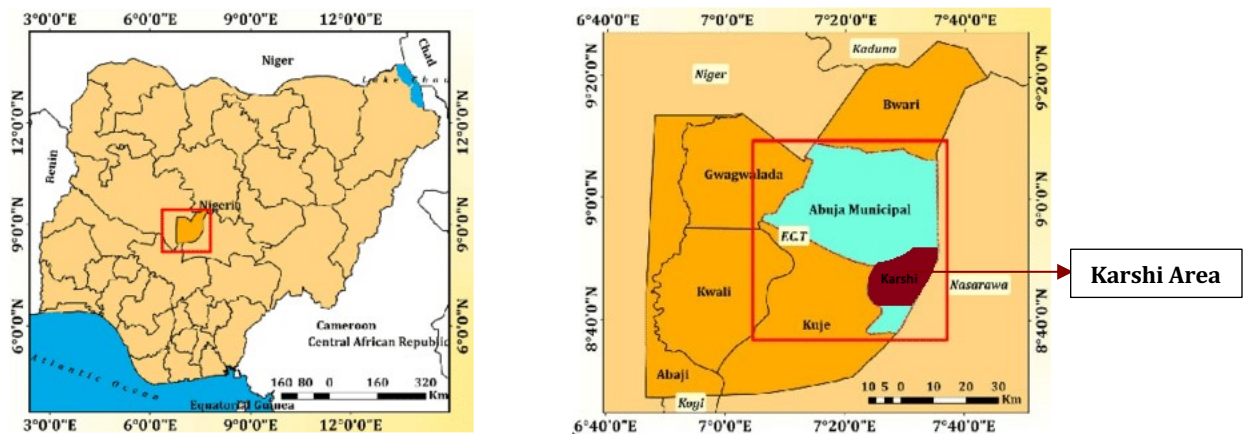


Figure 1: Map of Karshi Area in Abuja

Table 1: Monthly insolation in Karshi, Abuja

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Peak Sun Hours (kWh/m ²)	5.83	6.05	6.08	5.82	5.42	4.94	4.45	4.22	4.59	5.08	5.77	5.75

3.2. Design of proposed solar-powered water pumping system for irrigation

The proposed solar-powered pumping system's design is described in the following subsections.

3.2.1. Analysis of the System

During the dry season, demand for water is highest. The design month for the system will be August, which is the month with the lowest insolation (Table 1). Most farmers in this community practice crop rotation, and tomatoes are among the crops that generate the highest income during the dry season. Groundnuts and vegetables (non-water-intensive crops) are usually rotated with crops because these are the most commonly grown crops

during the rainy season. So, these crops (tomatoes, groundnuts and vegetables) were chosen for this study. The water needed for irrigation is based on the water requirements for these crops. Since pump capacity is selected based on the water demand of the irrigation system, factors such as crop water requirements, field size, and desired flow rates are considered to ensure adequate water supply for efficient irrigation (Bawa et al., 2023). The amount of water required for the selected crops are: groundnut: 7 m³/day, every fourth day; vegetables: 2 m³/day, every fourth day; tomatoes: 9 m³/day, every third day. Consequently, the irrigation system design will focus on tomatoes, which need the most water. The water source for this system is a well and a submersible pump with a 9.5 meters dynamic head was chosen for the design. The assumption is that the system's water tank should be able to hold three days of water needs. Since insolation may be lower than expected, the system will be designed to pump all the required water in one day. For tomatoes this means pumping 9 m³ in one day and using it when needed on the third day. Thus, the system requires a 10 m³ tank. This system will use a plastic tank measuring 2.5 m tall and 10 m³ in volume with inflow at the top and discharge at the bottom. A DC pump that can lift 10,000 litres per day is the recommended choice in this design.

4. Sizing of the Proposed System using HOMER

As stated in the preceding paragraph, an optimization program known as HOMER was used as a modeling tool and for data simulation and analysis. To model this situation in HOMER, the following were inputted in the deferrable load model.

1. The rated power of the pump, which is 0.26 kW
2. The storage capacity of the tank is 1.09 kWh
3. The peak load is 0.13 kW
4. The minimum load ratio is 72%.

The detailed load mapping in the deferrable load in HOMER can be seen in figure 2

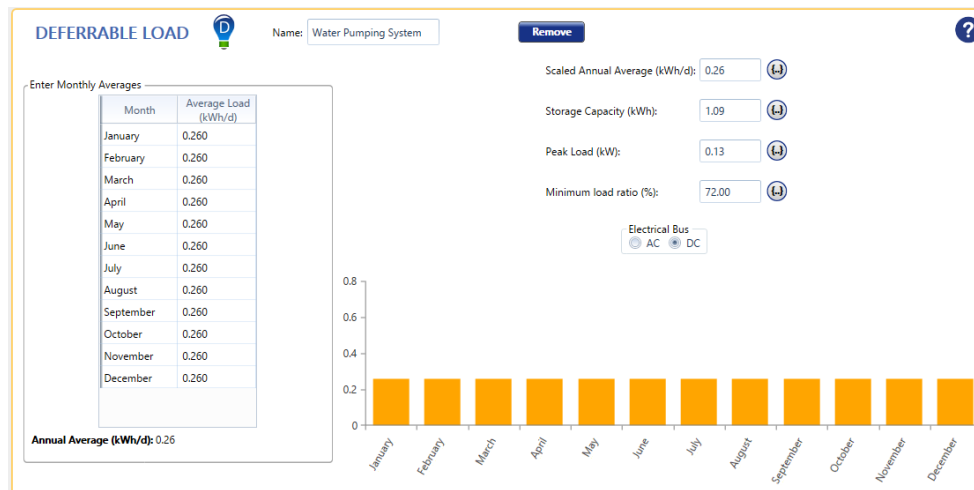


Figure 2: Load Mapping in HOMER.

The water pump required 0.13 kW of power to run at 100% of that power to operate as shown in the schematic diagram in Figure 3 and that of the diesel generator in Figure 4.

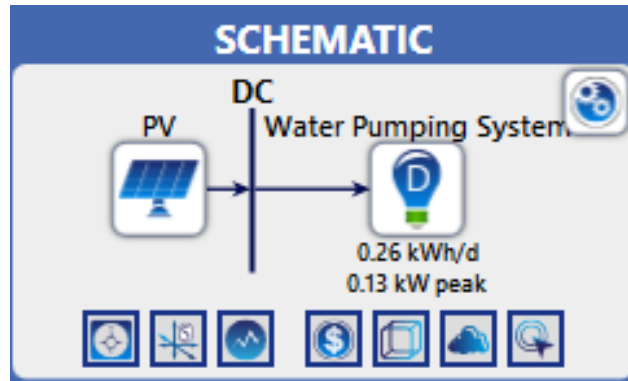


Figure 3: Schematic diagram of the designed PV system for water pumping application

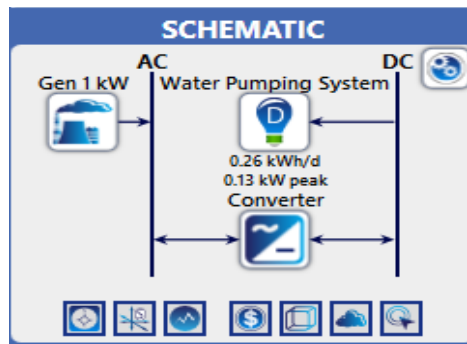


Figure 4: Schematic diagram of the diesel power system for water pumping application

5. Results and Discussion

The simulation details of the proposed PV system and base (diesel) system are presented in Table 2.

Table 2: Simulation details of the proposed and base system component

Quantity	PV System		Diesel System Component			
	PV		Diesel		Converter	
	Value	Units	Value	Units	Value	Units
Capacity	0.400	kW	0.500	kW	0.130	kW
Mean Output	0.0711	kW	0.137	kW	0.0107	kW
Minimum Output	0	kW	0.137	kW	0	kW
Maximum Output	0.379	kW	0.137	kW	0.130	kW
Capacity Factor	17.8	%	2.26	%	8.24	%
Hours of Operation	4,380	hrs./yr.	722	hrs./yr.	722	hrs./yr.
Number of Starts			722	Starts/yr.		
Electrical Production	623	kWh/yr.	98.8	kWh/yr.		
Fuel Consumption	0	L	38.9	L		
Energy In					98.8	kWh/yr.
Energy Out					93.9	kWh/yr.
Losses					4.9	kWh/yr.

Technical Analysis

The model (Figure 5) demonstrated that the system's energy requirement (the pumping load) in the designed PV system for water pumping application was satisfied by 0.40 kW of total PV capacity. A total annual energy output of 623 kWh has been generated by photovoltaics, maintaining continuous operation for 4,380 hours/year, and excess electricity of 528 kWh/year, as shown in Figure 5. This thereby supplies the peak load demand for the submersible pump to operate. The results obtained clearly show that the designed solar-powered water pumping system was able to produce the energy required for irrigation. A deferrable load, such as a water pump, is a flexible electrical device that requires a specific amount of energy over a period but does not need to operate at a specific time. Consequently, it is essential to design the system in a way that prevents the water tank from running dry. The reason for the high excess energy (528 kWh/year) is because of the absence of an energy storage system to store the generated energy for later use. The addition of a storage battery to the energy system will enable the farmer to utilize the surplus energy to operate his farmhouse at night.



Figure 5: PV electrical summary of the designed PV system for water pumping application

Compare Economics

From the compared economics in figure 6, it was observed that the capital cost of the proposed solar-powered water pumping system was 1.71 times that of the conventional diesel pumping system. Nevertheless, after 20 years of operation, the life cycle cost of the conventional diesel system would be 8.97 times that of the solar-powered water pumping system. The chart shows that the PV system has a large initial capital cost but a negligible operating cost (due to minimal maintenance and no fuel costs), while the diesel system has a relatively low initial capital cost and a significantly higher maintenance and operating cost. The findings of this analysis are in concordance with those from other studies regarding the economic viability of solar pumping systems compared with diesel pumping systems. The payback period for the solar-powered water pumping system is calculated by dividing the initial investment by the annual net savings. Therefore, it will take this system approximately 0.67 years to recover its initial costs. This demonstrates that the system is both cost-effective and sustainable in the long term.

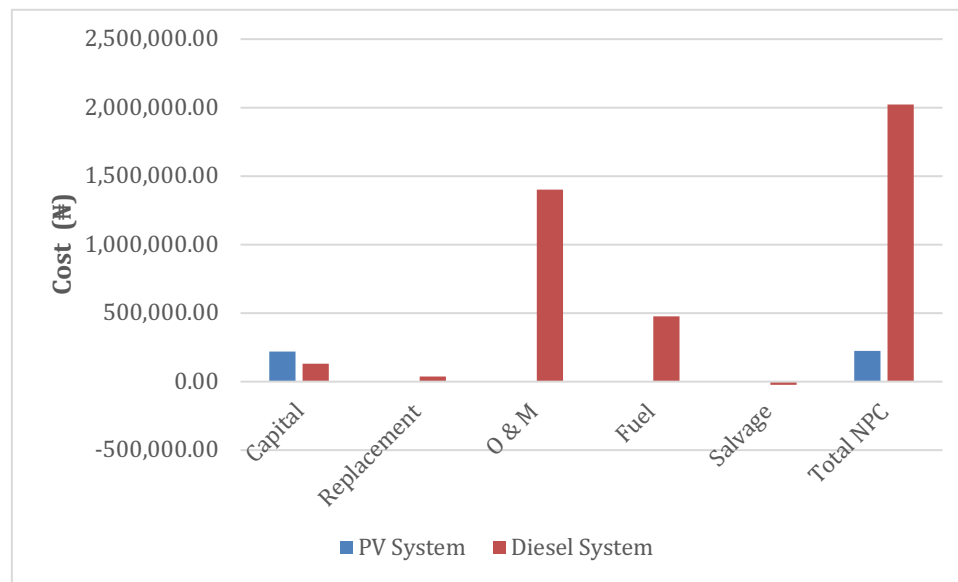


Figure 6: Cost summary of the proposed (PV system) and base (Diesel) system

6. Conclusion

In many parts of the Karshi community, most irrigation systems are fossil fuel-based, whereas a solar water pumping system can significantly mitigate greenhouse gas emissions while maintaining low operation and maintenance costs. To transition from this conventional pumping system to the off-grid solar system, this study presents a designed solar-powered water pumping system for irrigation and demonstrates a commitment to sustainability by reducing financial waste and adverse environmental impacts. A detailed design and cost analysis for a solar water pumping system and conventional diesel pumping system was performed in HOMER. From the economic point of view, it was observed that the diesel engine-powered pumping system is typically characterized by a lower first cost but very high operation and maintenance costs, while the solar-powered pumping system has a high initial cost but low maintenance and operation costs, which made the proposed solar-powered water pumping system more economical than the diesel engine-powered pumping system. Furthermore, from the environmental point of view, it was discovered that implementing the solar system could prevent the release of 102 kg of CO₂ per year, and this would lead to reduced consumption of diesel fuel by 38.9 L. This reduction in emissions highlights the potential for PV systems to mitigate the environmental impact of agricultural water pumping operations. This research work concluded that solar-powered pumping systems are a viable solution when compared with diesel-powered pumping systems and suggested that the utilization of solar energy for pumping water should be encouraged in Nigeria.

Conflict of Interest

No conflicts of interest.

Funding Statement

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